



ScamShield AI

Digital Public Safety Intelligence Platform
Technical & Executive Specification

1. Executive Summary & Problem Analysis

Digital fraud in India has evolved into an organized criminal enterprise. The Indian Cyber Crime Coordination Centre (I4C) reported over 694,000 cybercrime complaints in 2024, representing financial losses exceeding ₹11,333 Crore (\$1.35 Billion USD) in the first six months alone.

Among all fraud vectors, the 'Digital Arrest' scam is the most insidious. Perpetrators impersonate officials from the Central Bureau of Investigation (CBI), Narcotics Control Bureau (NCB), Enforcement Directorate (ED), or State Police departments. Victims are coerced into video calls, told that their identity or bank accounts are linked to illegal activities, and forced into continuous monitoring under threat of immediate physical arrest.

Quantitative Fraud Impact Matrix

Fraud Vector	Annual Case Count	Avg Loss per Victim	Psychological Mechanism
Digital Arrest	7,061 (2024)	₹15.5 Lakhs	Coercive fear & authority pressure
UPI / Payment Trap	312,000 (2024)	₹42,000	Speed urgency & cognitive overload
Phishing / Impersonation	185,000 (2024)	₹88,000	Trust exploitation & deceptive URLs
Investment / Stock Scams	98,000 (2024)	₹4.2 Lakhs	Greed amplification & fake returns

2. Biological & Societal Context of Digital Arrest Scams

The success of digital arrest scams relies on exploiting human stress neurobiology. When a scammer claims to be a police officer threatening immediate imprisonment:

- 1. The victim's amygdala triggers a sympathetic nervous system fight-or-flight response.
- 2. Adrenaline and cortisol surge, suppressing executive functioning in the prefrontal cortex.
- 3. Rational scrutiny drops significantly, making the victim highly susceptible to instructions such as transferring funds to 'government verification accounts'.

ScamShield AI acts as an artificial prefrontal cortex guardrail. By displaying objective, mathematical risk meters and visual alerts, the platform interrupts the panic loop and forces analytical cognition to re-engage.

3. System Architecture & Data Engineering

ScamShield AI is engineered as a decoupled, multi-tiered architecture designed for low latency, high throughput, and zero single-point-of-failure operation.

- Frontend Layer: Built with Next.js 14 (App Router), TypeScript, and Tailwind CSS with custom glassmorphism styles.
- Backend Layer: Built with Python 3 and FastAPI, featuring asynchronous processing, rate limiting, and security middleware.
- Real-Time Communication Layer: Bi-directional WebSocket channels for streaming speech transcripts and real-time risk scores.
- AI Engine Layer: Multi-model pipeline featuring Google Gemini (primary), Hugging Face (secondary), and a local deterministic heuristic engine (fallback).

4. AI Engine & Real-Time NLP Pipeline

The ScamAnalyzer service implements a three-tier fallback chain. If cloud APIs are unreachable, the local heuristic engine scores inputs based on four domain risk vectors:

- Authority Impersonation Vector (Weight 30%): Matches phrases like CBI, Police, Customs, Supreme Court.
- Coercion & Legal Threats Vector (Weight 35%): Matches phrases like digital arrest, non-bailable warrant, Section 420.
- Financial Demand Vector (Weight 25%): Matches phrases like verification account, UPI deposit, security fee.
- Urgency & Secrecy Vector (Weight 10%): Matches phrases like immediate, keep quiet, do not disconnect.

5. Graph Theory & Threat Intelligence Mathematics

The Threat Intelligence Dashboard incorporates an interactive force-directed graph to map fraud networks. Node coordinates are dynamically calculated using Coulomb repulsion and Hooke spring attraction physics:

Repulsive Force (Coulomb's Law): $F_r(i, j) = - (k_r / ||r_{ij}||^2) * \hat{r}_{ij}$

Attractive Force (Hooke's Law): $F_a(i, j) = k_a * (||r_{ij}|| - l_0) * \hat{r}_{ij}$

Where $k_r = 5000$ is the repulsion constant, $k_a = 0.05$ is spring stiffness, and $l_0 = 100$ is edge rest length.

6. Gamification & Behavior Change Engine

To transform passive users into active fraud detectors, ScamShield AI implements an incentive mechanics loop:

- Scan Message: +10 Points
- Share Fraud Alert on WhatsApp: +15 Points
- Quiz Correct Answer: +20 Points
- Complete Full Quiz: +50 Points
- Submit Verified Report to Authorities: +25 Points

7. Security Hardening & Threat Model

ScamShield AI enforces strict HTTP security headers (X-Frame-Options DENY, X-Content-Type-Options nosniff, Permissions-Policy, HSTS), origin-restricted CORS policies, per-IP rate limiting (SlowAPI), and short-lived JWT tokens for WebSocket connections.